

15.2 Kinetic Theory

Question Paper

Course	CIE A Level Physics
Section	15. Ideal Gases
Topic	15.2 Kinetic Theory
Difficulty	Easy

Time allowed: 40

Score: /31

Percentage: /100

Question 1a

The kinetic theory of gasses is based upon a number of assumptions.

Place ticks (✓) next to the correct assumptions in Table 1.1.

Table 1.1

Possible statements of the kinetic theory of gases	Place a tick (✓) here
The time of a collision is equal to the time between collisions	
The molecules are mostly stationary but move upon collision with another molecule	
Molecules of gas behave as identical, hard, perfectly elastic spheres	
The volume of molecules is significant compared to the volume of the container	
The time of a collision is negligible compared to the time between collisions	
There are no forces of attraction or repulsion between the molecules	
The molecules are in continuous random motion	
The molecules all have mass, so are attracted towards each other	
The volume of the molecules is negligible compared to the volume of the container	

[5 marks]

Question 1b

Draw a circle around the correct symbol for the mean square speed of molecules in an ideal gas.

$\sqrt{c}$ $\sqrt{\langle c \rangle}$ $\sqrt{\langle c^2 \rangle}$ c $\langle c^2 \rangle$
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[1 mark]**Question 1c**

State the units for the mean square speed of the molecules in an ideal gas.

[1 mark]**Question 1d**

Calculate the average root mean square speed of the particles travelling at the following velocities:

$$350 \text{ m s}^{-1} \quad -356 \text{ m s}^{-1} \quad 346 \text{ m s}^{-1} \quad -351 \text{ m s}^{-1} \quad 339 \text{ m s}^{-1}$$

[3 marks]**Question 2a**

State the kinetic theory of gases equation.

[2 marks]

Question 2b

A tank of volume 20 m^3 contains 6.0 moles of an ideal monatomic gas. The temperature of the gas is 38°C .

Calculate the average kinetic energy of the particles in the gas.

[4 marks]**Question 2c**

The following paragraph is an explanation of the relationship between the temperature of a gas and its pressure in terms of the kinetic model of an ideal gas.

Use the words in the box below to complete the sentences in the paragraph. They can be used once, more than once or not at all.

higher	lower	more	kinetic energy	force	momentum	pressure	less
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- i. A _____ temperature implies _____ average speed and therefore higher _____.
- ii. This causes an increase in the _____ transferred to the walls from _____ frequent collisions.
- iii. This increased _____ per collision leads to an increased _____.

[3 marks]

Question 2d

State the type of collision assumed between the molecules and the walls of the container in the kinetic theory of gases.

[1 mark]

Question 3a

This question is about the derivation of the kinetic theory of gases equation.

The derivation is based on a particle inside a cube, as shown in Fig. 1.1.

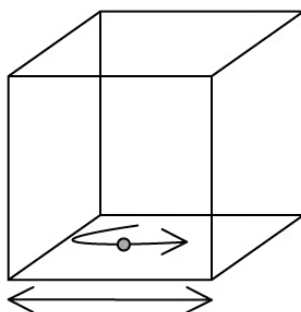


Fig. 1.1

Identify the following by adding the indicated letter to the correct arrow on the diagram:

(i)

The length l of the side of the cube

[1]

(ii)

The speed c of the molecule.

[1]

[2 marks]

Question 3b

Step two of the derivation of the kinetic theory of gases is explained by the statements in Table 1.1.

Place ticks (✓) next to the phrases that correctly explain step two.

Table 1.1

Some phrases about the derivation of the kinetic theory of gases	Place a tick (✓) here
c is the specific heat capacity of the particles of the gas	
c is not the speed of light	
Time between particle collisions with the container walls = $\frac{2 \times \text{length of cube}}{\text{speed of molecule}}$	
Time between particle collisions = $\frac{\text{distance}}{\text{speed}}$	
Time to travel length of cube = $\text{length of cube} \times \text{speed of molecule}$	

[3 marks]

Question 3c

Step four of the derivation of the kinetic theory of gases equation is explained in the sentences below.

Identify the correct words from the box to complete the sentences. You can use each word once, more than once or not at all.

velocity mean speed number amount area force exerted molecule

- Pressure is defined as the force per unit _____. This is the pressure exerted by one _____. The _____ of one wall is P .
- The gas contains a large _____ of molecules. Each molecule has a different _____ and they all contribute to the pressure.
- The _____ squared speed of c^2 is written with left and right-angled brackets $\langle c^2 \rangle$.

[3 marks]**Question 3d**

For the final step of the derivation of the kinetic theory of gases equation, state

(i)
the number of dimensions the molecules are moving in

[1]

(ii)
the volume of the cube in terms of its side length

[1]

(iii)
the equation for the density of the cube.

[1]

[3 marks]



Model Answers: Easy

1a

(a) The correct assumptions of kinetic theory are:

Possible statements of the kinetic theory of gases	Place a tick (✓) here
The time of a collision is equal to the time between collisions	
The molecules are mostly stationary but move upon collision with another molecule	
Molecules of gas behave as identical, hard, perfectly elastic spheres	✓ [1 mark]
The volume of molecules is significant compared to the volume of the container	
The time of a collision is negligible compared to the time between collisions	✓ [1 mark]
There are no forces of attraction or repulsion between the molecules	✓ [1 mark]
The molecules are in continuous random motion	✓ [1 mark]
The molecules all have mass, so are attracted towards each other	
The volume of the molecules is negligible compared to the volume of the container	✓ [1 mark]

- *One mark for each correct statement*

[Total: 5 marks]

1b

(b) The symbol for the mean square speed of molecules in an ideal gas is:

- $\langle c^2 \rangle$ [1 mark]

[Total: 1 mark]

1c

(c) The units for the mean square speed of the molecules in an ideal gas are:

- $\text{m}^2 \text{s}^{-2}$ [1 mark]

[Total: 1 mark]

1d

(d) Calculate the average root mean square speed of the particles travelling at the following velocities:

Calculate the sum of the square of the velocities:

- $\sum c^2 = 350^2 + (-356)^2 + 346^2 + (-351)^2 + 339^2 = 607\,074$ [1 mark]

Calculate the mean of the square of the velocities:

- $\langle c^2 \rangle = \frac{607\,074}{5} = 121\,415$ [1 mark]

Calculate the root of the mean of the square of the velocities:

- $\sqrt{\langle c^2 \rangle} = \sqrt{121\,415} = 348 \text{ m s}^{-1}$ [1 mark]

[Total: 3 marks]

2a

(a) The kinetic theory of gases equation is:

- $pV = [1 \text{ mark}] \frac{1}{3} Nm \langle c^2 \rangle [1 \text{ mark}]$

[Total: 2 marks]

2b

(b) Calculate the average kinetic energy of the particles in the gas:

List the known quantities:

- Temperature, $T = 38^\circ\text{C}$
- Boltzmann's constant, $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$

Convert temperature T from $^\circ\text{C}$ to K:

- $T(\text{K}) = T(^{\circ}\text{C}) + 273.15$
- $T = 38 + 273.15 = 311.15 \text{ K} [1 \text{ mark}]$

State the equation for average kinetic energy:

- $\overline{E}_k = \frac{3}{2} kT = \frac{3}{2} \frac{R}{N_A} T [1 \text{ mark}]$

Substitute values into average kinetic energy equation:

- $\overline{E}_k = \frac{3}{2} \times (1.38 \times 10^{-23}) \times 311.15 [1 \text{ mark}]$
- $\overline{E}_k = 6.44 \times 10^{-21} \text{ J} [1 \text{ mark}]$

[Total: 4 marks]

It is good to keep in mind that the **average kinetic energy equation** is the kinetic energy of **one** molecule in the gas. This is why the number of molecules hasn't been calculated from the number of moles – both the moles and volume are just a **red herring** in this question!

Remember that temperature should always be in **kelvin (K)** in this topic.

2c

(c) Correct sentences that would score 3 marks should look like this:

- A **higher** temperature implies **higher** average speed and therefore higher **kinetic energy**; [1 mark]
- This causes an increase in the **momentum** transferred to the walls from **more** frequent collisions; [1 mark]
- This increase in **force** per collision leads to an increase in **pressure**; [1 mark]

One mark for each correct sentence

[Total: 3 marks]

The reason that it is technically the increased **force** per collision that leads to an increased pressure, is due to the definition of pressure. Pressure is the 'force per unit area' ($p = \frac{F}{A}$) and the area of the collisions remain constant. Remember, this has nothing to do with the ideal gas equation $pV = nRT$, and any reference to this will be ignored by the examiner.

2d

(d) The collision assumed between the molecules and the walls of the container in the kinetic theory of gases is:

- Elastic; [1 mark]

[Total: 1 mark]

3a

(a)

A diagram that would score 1 mark should look like this:

One mark for each correct sentence

[Total: 3 marks]

The reason that it is technically the increased **force** per collision that leads to an increased pressure, is due to the definition of pressure. Pressure is the 'force per unit area' ($p = \frac{F}{A}$) and the area of the collisions remain constant. Remember, this has nothing to do with the ideal gas equation $pV = nRT$, and any reference to this will be ignored by the examiner.

2d

(d) The collision assumed between the molecules and the walls of the container in the kinetic theory of gases is:

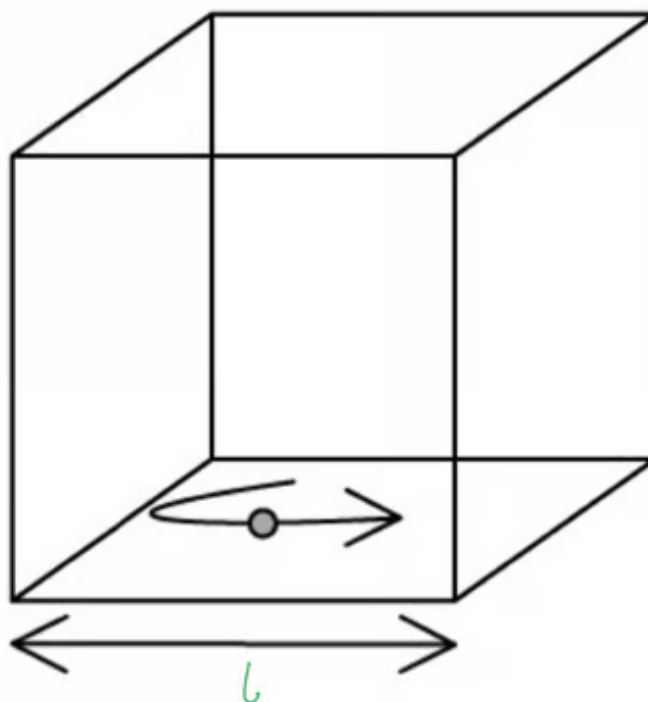
- Elastic; [1 mark]

[Total: 1 mark]

3a

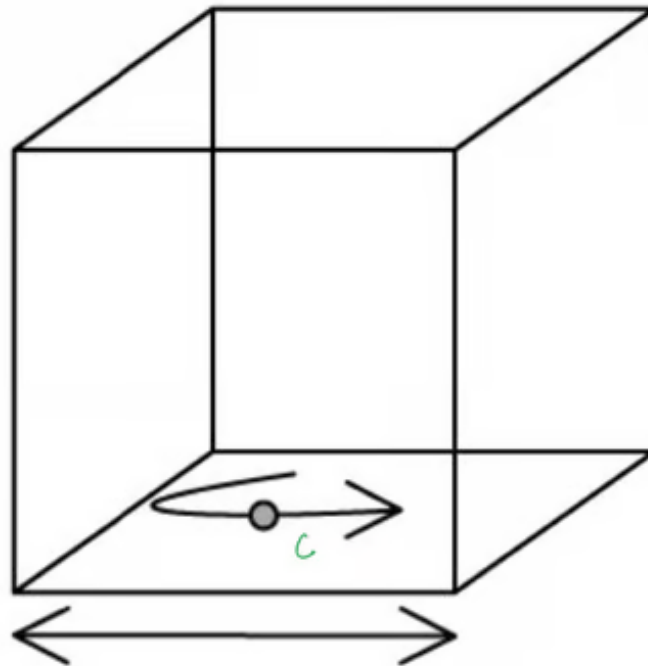
(a)

A diagram that would score 1 mark should look like this:



- Horizontal arrow below the cube labelled with the letter l ; [1 mark]

A diagram that would score 1 mark should look like this:



- Curved arrow inside the cube pointing to the right labelled with the letter c ; [1 mark]

[Total: 2 marks]

3b

(b) A correctly marked table that would score 3 marks should look like this:

Some phrases about the derivation of the kinetic theory of gases	Place a tick (✓) here
c is the specific heat capacity of the particles of the gas	
c is not the speed of light	✓ [1 mark]
Time between particle collisions with the container walls = $\frac{2 \times \text{length of cube}}{\text{speed of molecule}}$	✓ [1 mark]
Time between particle collisions = $\frac{\text{distance}}{\text{speed}}$	✓ [1 mark]

Time to travel length of cube = $\text{length of cube} \times \text{speed of molecule}$	
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- *One mark for each correct statement ticked*

[Total: 3 marks]

3c

(c) Sentences that correctly explain step four of the derivation should look like this:

- Pressure is defined as the force per unit **area**. This is the pressure exerted by one **molecule**. The **area** of one wall is l^2 ; [1 mark]
- The gas contains a large **number** of molecules. Each molecule has a different **velocity** and they all contribute to the pressure; [1 mark]
- The **mean** squared speed of c^2 is written with left and right-angled brackets $\langle c^2 \rangle$; [1 mark]

- *One mark awarded for each correct sentence*

[Total: 3 marks]

3d

(d)

(i) The molecules move in:

- 3-dimensions; [1 mark]

(ji) The volume of the cube in terms of its side length is:

- length cubed or l^3 ; [1 mark]

(iii) The equation for the density of the cube is:

- $density = \frac{mass}{volume}$ [1 mark]

OR

- $density = \frac{Nm}{V}$ [1 mark]

[Total: 3 marks]

